Predicting Employee Attrition Using Machine Learning

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Business Problem and Objective

- Employee turnover leads to high recruitment and training costs
- Traditional exit interviews are reactive, not predictive
- **Objective:** Predict employee attrition using ML models
- **Goal:** Enable proactive retention strategies and reduce turnover

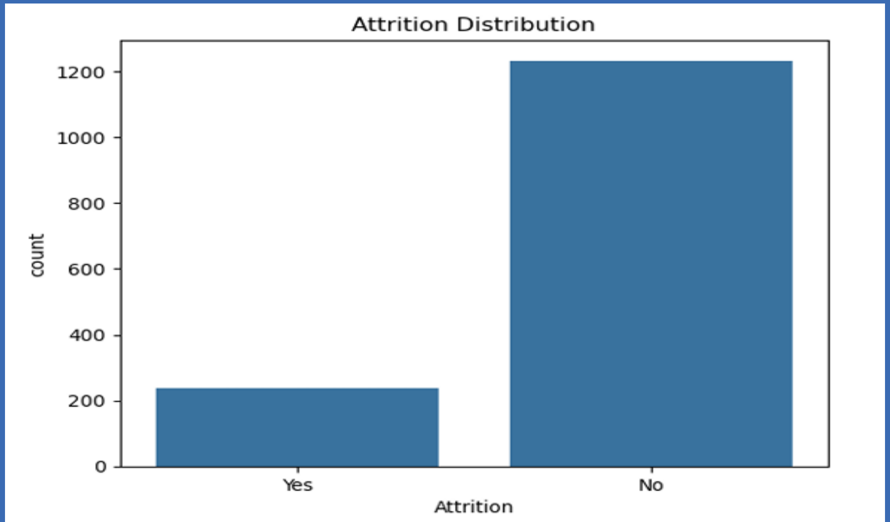
Dataset and Preprocessing

- Source: Kaggle – IBM HR Analytics Employee Attrition dataset
- 1,470 records, 35 features
- Key variables: Age, Job Role, Monthly Income, OverTime, Attrition
- Preprocessing:
  - *Encoded categorical features*
  - *Scaled numerical data*
  - *Handled class imbalance (Attrition: Yes vs No)*

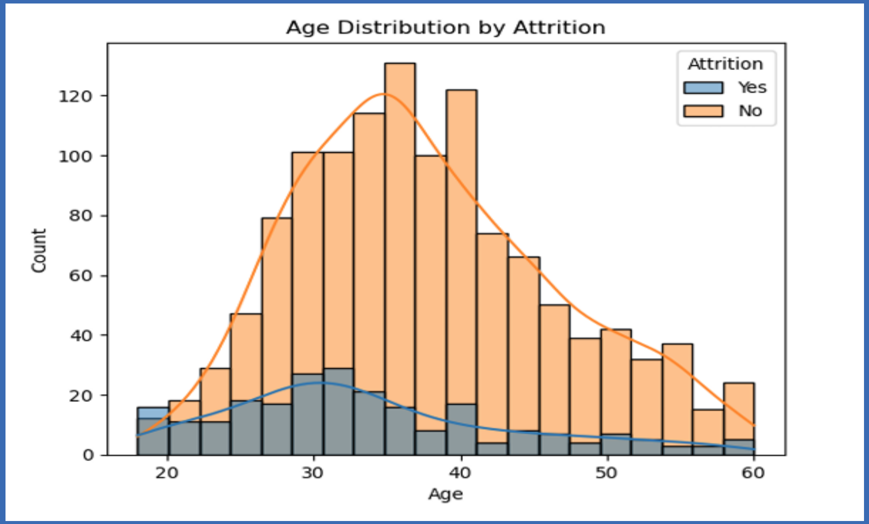
Exploratory Data Analysis (EDA)

- OverTime and Job Role show strong relation to attrition
- Employees with OverTime are more likely to leave
- Younger and lower-income employees show higher attrition
- Some roles like Sales and HR see more turnover
- Class imbalance: Only ~16% attrition rate

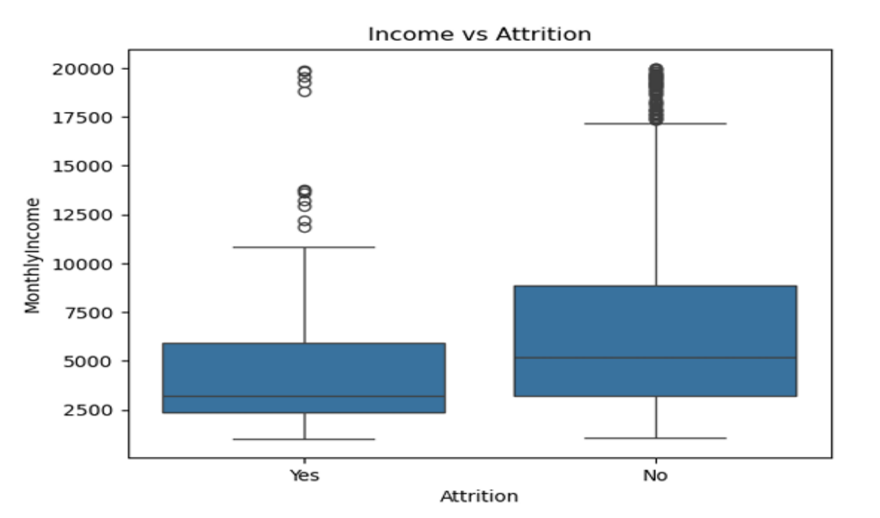
EDA Visuals



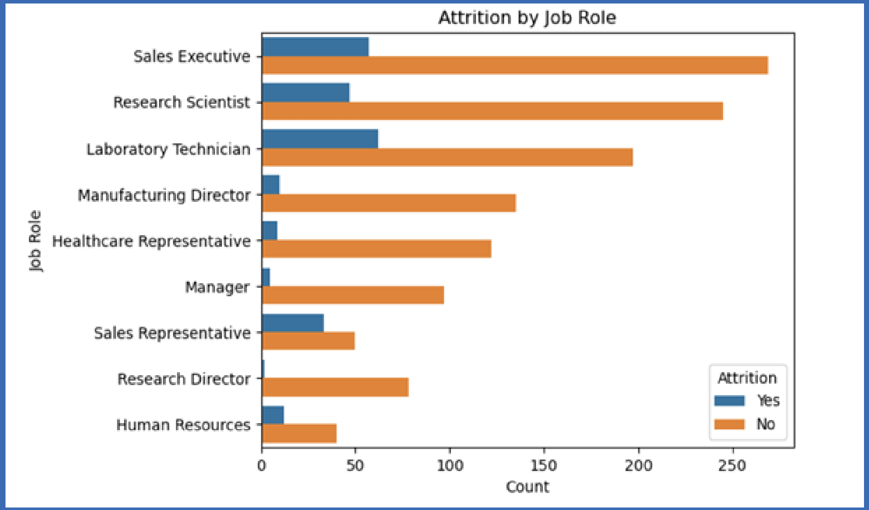
Attrition Distribution
count
1200
1000
800
600
400
200
0
Yes
No
Attrition



Age Distribution by Attrition
Attrition
Yes
No
Count
120
100
80
60
40
20
0
20
30
40
50
60
Age



Income vs Attrition
MonthlyIncome
20000
17500
15000
12500
10000
7500
5000
2500
Yes
No
Attrition



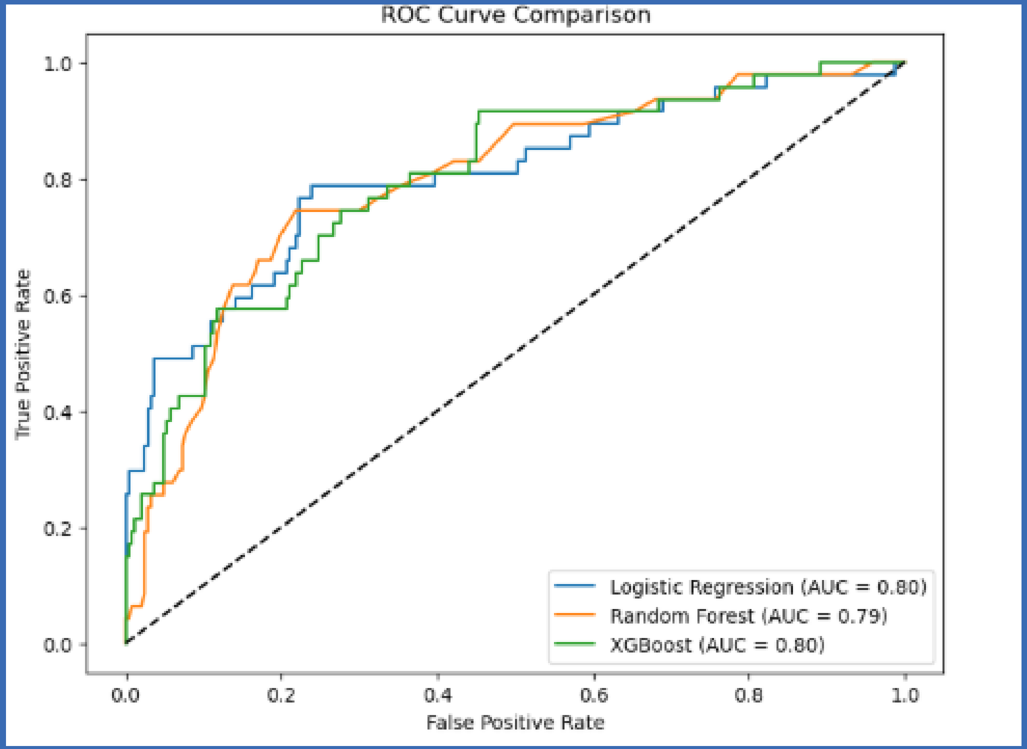
Attrition by Job Role
Job Role
Sales Executive
Research Scientist
Laboratory Technician
Manufacturing Director
Healthcare Representative
Manager
Sales Representative
Research Director
Human Resources
Attrition
Yes
No
0
50
100
150
200
250
Count

Modeling Approach

- Target: Predict employee attrition (Yes/No)
- Addressed class imbalance with SMOTE
- Used Logistic Regression, Random Forest, and XGBoost
- Compared model performance on balanced data
- Logistic Regression showed best recall for attrition class

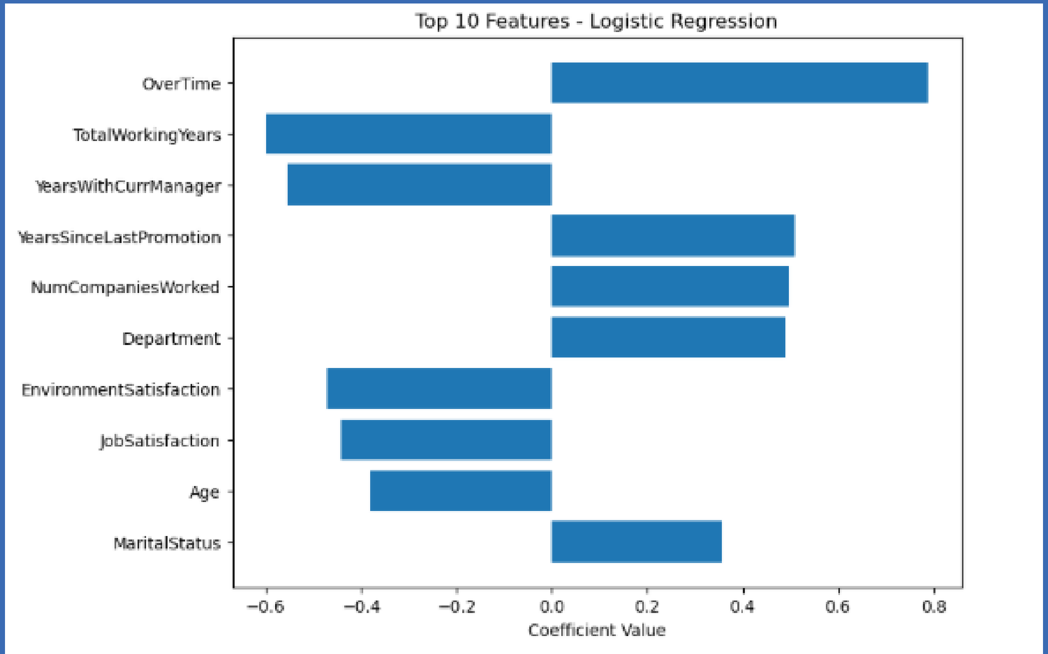
Model Results & Evaluation

- Logistic Regression: Best recall for attrition (77%)
- Random Forest & XGBoost: Higher overall accuracy (~85%)
- Precision-Recall tradeoff: Logistic Regression has more false positives
- ROC-AUC scores similar (~0.80) across models
- Confusion Matrices show Logistic Regression better identifies attrition cases



Feature Importance (Logistic Regression)

- OverTime is the strongest positive predictor of attrition.
- Longer Years with Current Manager reduces attrition risk.
- Higher Job Satisfaction and Environment Satisfaction lower likelihood to leave.
- Older employees are less likely to leave (Age negatively associated).



Implementation and Use Cases

- Pilot the model with current employee data to validate predictions
- Integrate into HR systems for real-time attrition risk monitoring
- Train HR teams on interpreting model results and taking action
- Use predictions to proactively manage overtime and job satisfaction
- Target mentorship and engagement programs for high-risk employees
- Regularly update and retrain the model for accuracy

Limitations and Ethical Concerns

- Dataset limited to one company, missing external factors
- Possible bias in self-reported features
- Model may not generalize across industries
- Ensure fairness and avoid discrimination
- Protect employee privacy and secure data
- Use predictions to support employees, not penalize
- Maintain transparency to build trust

Conclusion & Final Thoughts

- Successfully developed models predicting employee attrition
- Logistic Regression best identifies at-risk employees
- Key factors: OverTime, manager relationship, job satisfaction
- Enables proactive HR interventions to reduce turnover
- Future work: refine models, add data, monitor regularly

References

- IBM HR Analytics Employee Attrition & Performance. (2017, March 31). Kaggle. https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
- scikit-learn: machine learning in Python — scikit-learn 1.7.0 documentation. (n.d.). https://scikit-learn.org/stable/
- Preet, A. (2024, November 11). Employee Attrition Prediction – A comprehensive guide. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2021/11/employee-attrition-prediction-a-comprehensive-guide/